Anant Verma

New Delhi, India | +91-8368570080 | anantverma163@gmail.com | linkedin.com/in/anant-verma- | github.com/anant7852
Portfolio

Summary

Data engineer with applied ML and LLM experience: Airflow-orchestrated production pipelines, gradient boosting demand and price forecasting, a grounded biomedical RAG system with a measured evaluation harness (96% recall, 100% grounding), and a text-to-SQL agent for a production warehouse. Strong foundation in Python, SQL, workflow orchestration, and production data pipelines at a Series-A mobility startup.

Technical Skills

ML & GenAI: Scikit-learn, gradient boosting, feature engineering, RAG, embeddings, hybrid retrieval (dense + BM25), LLM agents and tool calling (Agno), Gemini API, Groq, sqlglot, prompt engineering
Languages & Databases: Python, SQL, PostgreSQL, Amazon Redshift, MongoDB (aggregation pipelines)
Analytics: Pandas, NumPy, Matplotlib, EDA, KPI design (GMV, CM1, ASP), statistics, Excel, Google Sheets
Data Engineering: Apache Airflow (DAG authoring, scheduling, Mongo→Postgres sync), ETL/ELT pipelines, Flask REST APIs, Redash API, Google Apps Script, data quality & reconciliation
Tools: Git, Bitbucket, pgAdmin, Jupyter, Next.js, Vercel

Projects

Evidentia – Grounded Biomedical RAG System RAG, Hybrid Retrieval, Gemini, Next.js | Live | Code
- Built a retrieval-augmented Q&A system over 400 CORD-19 vaccine-efficacy papers using hybrid dense + BM25 retrieval, query decomposition, and citation-grounded generation; deployed serverless on Vercel.
- Designed a 35-question evaluation harness: 96% retrieval recall, 100% answer grounding, 71% correct abstention on unanswerable queries.

Yield Copilot – In-Browser Demand & Price Forecasting Gradient Boosting, Scikit-learn, Next.js | Live | Code
- Trained gradient boosting models for seat demand and fare optimisation; exported model trees to JSON for zero-latency client-side inference, verified to match Scikit-learn predictions.
- Built an interactive revenue-vs-price explorer that recomputes forecasts live and surfaces the revenue-maximising fare per departure.

Zingbus SQL Agent – Text-to-SQL for the Production Warehouse Agno, Groq LLM, sqlglot, PostgreSQL
- Built an LLM agent that turns plain-English metric requests into PostgreSQL for a 71-table, 1,415-column warehouse, grounded through 8 tools for schema lookup and retrieval over a 24-section metric-definition guide.
- Added offline SQL validation with sqlglot, checking every table and column against the parsed schema to catch hallucinated columns, unknown tables, ambiguous references, and unquoted identifiers before output.

Experience

Zingbus Gurugram, India
Data Engineer *Aug 2026 – Present*
Data Analyst Intern *Feb 2026 – Jul 2026*
- Build and maintain Apache Airflow DAGs orchestrating MongoDB-to-PostgreSQL syncs and third-party API ingestion (EV operator contracts, chatbot session logs) on scheduled master and incremental chains; manage warehouse tables, schemas, and queries in pgAdmin.
- Built and maintain production ETL pipelines in Python and SQL computing trip-level financial KPIs (GMV, CM1, ASP, OTA commission) across 4+ OTA partners, sourcing from MongoDB, PostgreSQL, and Redash and serving results through Flask APIs.
- Developed gradient boosting models forecasting seat demand and ASP across bus routes, engineering seasonality, lag, and route-level aggregate features; currently in validation ahead of rollout to the ops team.
- Audited the daily sales report pipeline and fixed silent data errors: an INR 8.37 lakh cancellation-charge double-count, 2,353 trips misclassified by service type, and a merge fan-out corrupting 5,324 rows.
- Built an EV fleet analytics pipeline joining BMS and JBM telematics, charging records, and RFID mappings to attribute battery state of charge and kilometres to trips; deployed as Flask endpoints.
- Built the operator revenue-share payout engine and a 47-column trip-level API, parsing per-trip contracts, sliding-scale payout tiers, and rolling bus-pair cycles; traced every rupee from paid amount to payout across thousands of trips.

Leadership

180 Degrees Consulting, NSUT Jan 2025 – May 2026
Consulting Director *Delhi, India*
- EVA2Z (EV startup): led financial risk modelling, customer segmentation, and pricing optimisation yielding an 18% profit increase.
- PLAT Exam (ed-tech): built a data-driven go-to-market strategy from student performance data, reaching 5,000+ rural students.

Education

Netaji Subhas University of Technology (NSUT) 2022 – 2026
B.Tech, Biotechnology | CGPA: 8.12/10 *Delhi, India*